

# GENERAL THETA FOUNDATIONS I: COMPATIBILITY OF POSITIVE MEMORY AND ONLINE SIMPLEX SYNTHESIS

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**ABSTRACT.** We distinguish the separate positive realization of continuation arrays from their simultaneous realization by a finite causal machine. In the clocked atomic stochastic-row model, canonical affine sections yield a simultaneously rank-minimal machine whenever those sections are simplices; in particular, full-support arrays of cut rank at most two have no compatibility obstruction. We construct a full-support family of cut rank three whose two individual minimum state counts are both three, but whose exact Pareto-minimal profiles are  $(3, 4)$  and  $(4, 3)$ . Thus the compatibility obstruction occurs already at rank three and persists away from support boundaries. In the opposite direction, every additive stochastic encoder admits an online realization with no increase in its label alphabet. Consequently every simplex enclosing a contracted cube gives an online query encoder with the same number of states. This transfers all classical critical cube-simplex inclusions, including the non-Hadamard dimensions five and nine, to optimal online realizations. We also identify exact conditional query synthesis with classical worst-case random-access geometry at a checkpoint, and separate that equivalence from the cost of computing an encoder. Explicit rational tables, a finite fair-bit sampling ledger, and internal stochastic-language conventions complete the realization statements.

## 1. INTRODUCTION

A finite statistical response law may have a small positive factorization at every cut and nevertheless require a larger state space when those factorizations must be executed in succession. The difficulty is not linear dimension alone: the generators selected at one cut determine continuations which must be available at the next. Conversely, some families have enough additive structure that a static factorization already specifies a lossless online implementation. This paper proves both statements in one precise model.

Our first main result is the family in Theorem 3.3. An input with four possible values is followed by a five-valued command and a binary query. Only the final binary answer is released. Each of the two internal cuts separately has minimum state count three. No machine has three states at both cuts, even though all prescribed answer probabilities can be bounded between  $1/20$  and  $19/20$ . The profiles  $(3, 4)$  and  $(4, 3)$  are both realized explicitly. The proof excludes arbitrary stochastic encoders and decoders, not merely residual-state machines or a chosen architecture. It uses a forced affine slice at the first cut and a quantitative triangle constraint at the second.

The contrast is sharp in cut rank. Proposition 2.1 shows that the intersection of the normalized causal polytope with the affine span of the actual residuals is closed under normalized shifts. If these canonical sections are simplices, their vertices provide a simultaneously optimal machine (Theorem 2.2). All nonempty one-dimensional sections are intervals, so full-support cut ranks at

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most two cannot exhibit the preceding failure. This criterion uses an intrinsic polytope, not unknown transition coefficients. The invariant-cone and residual-realization mechanisms have classical antecedents [5, 10, 2]; the explicit rank-three obstruction is stated separately from them.

Our second main result is Theorem 4.1. A stochastic encoder on a product input alphabet which is additive in its input coordinates can be computed by a finite-state replacement process using exactly the same label alphabet. There is no retained coordinate selector or common random seed. An enclosing simplex has affine barycentric coordinates, so every static simplex encoder for the binary-query channel

$$(1) \quad \mathbb{P}(B = b \mid x, j) = \frac{1 + b\eta x_j}{2}, \quad x \in \{-1, 1\}^n, \quad j \in [n], \quad b \in \{-1, 1\},$$

has an online realization of the same peak. In particular, in every dimension and at every nonzero signal,

$$(2) \quad K_n(\eta) = n + 1 \iff W_n^{\text{fo}}(\eta) = n + 1 \iff W_n^{\text{ad}}(\eta) = n + 1.$$

The quantities on the right count all acquisition cuts;  $K_n$  constrains only the cut preceding the query. The proof of the first implication works for every enclosing simplex, not only orthogonal or Hadamard ones.

The static critical inequality  $\eta \leq 1/n$  is classical cube–simplex geometry. We state its relationship to the absorption index explicitly. Nevskii and Ukhalov’s critical simplices in dimensions five and nine [7] give, by the online theorem, six- and ten-state implementations at  $\eta = 1/5$  and  $\eta = 1/9$ . We display the rational vertices and the replacement distributions, so this consequence does not rely on a hypothetical static-to-dynamic transfer. Critical simplex matrices are also closed under a normalized tensor construction, giving further explicit online families without invoking unproved Hadamard existence. The static inclusions are not claimed as new.

A further distinction can be settled exactly. The minimum checkpoint alphabet for (1) equals that for classical private-randomness random access with worst-case success at least  $(1 + \eta)/2$ . This follows already from the geometric characterization and cube-containment argument of Kondo et al. [6, Theorem 14, Lemma 15]. We give the short normalization and exactification proof in Section 6. No extra persistent symmetrization seed is required. Therefore exact conditional rows do not themselves establish a different checkpoint exponent from classical random-access coding [1]. The online implementation, not that exponent, is the object of the present synthesis theorem.

| Notation                           | Meaning                                                 |
|------------------------------------|---------------------------------------------------------|
| $\mathcal{R}_t, \mathcal{P}_t$     | actual normalized residuals, all legal future arrays    |
| $L_t, r_t$                         | canonical affine section, affine dimension plus one     |
| $(K_1, K_2)$                       | simultaneous register bounds at two designated cuts     |
| $K_n(\eta)$                        | query-cut minimum, with unrestricted upstream memory    |
| $W_n^{\text{fo}}, W_n^{\text{ad}}$ | full acquisition peak: fixed or adaptive order          |
| $\xi(S)$                           | cube absorption by homothety about the simplex centroid |

A state count means the number of persistent labels. The primary model has exact atomic stochastic rows and an externally declared clock. Program descriptions and row-construction time are not part of that count. Section 7 gives a separate finite fair-bit implementation with its additional phase and sampling registers. A query is a current input, not a stored word; a model requiring a query buffer must charge it. These conventions are used consistently in both upper and lower bounds.

## 2. CANONICAL AFFINE SECTIONS OF A CAUSAL ARRAY

We use finite alphabets and a finite horizon. At a controller stage an action is chosen before its next report is received. A terminal answer is an action with a dummy following report. A complete normalized causal array  $C(\mathbf{a} \mid \mathbf{y})$  prescribes the conditional law of all actions for every externally supplied report word. Nonanticipation means that the marginal of an action prefix depends only on reports preceding that prefix. Intermediate buffers that must survive an update are included as charged cuts. The rows may depend on the declared time, but not on an unrecorded history, schedule variable or persistent random seed.

At a cut  $t$ , let  $\mathcal{P}_t$  be the polytope of normalized causal arrays on the remaining interface. For a controller prefix  $h$  whose probability, with past reports fixed, is positive, let  $R_h \in \mathcal{P}_t$  be its normalized future array. Remove duplicate rows to form the finite residual set  $\mathcal{R}_t$ . This is a property of  $C$ , not of a chosen realization. We require the full array, including unsupported environmental inputs; an almost-sure partial specification is a different completion problem. Set

$$(3) \quad A_t = \text{aff } \mathcal{R}_t, \quad L_t = A_t \cap \mathcal{P}_t, \quad r_t = \dim A_t + 1.$$

The slice  $L_t$  is a compact polytope of dimension  $r_t - 1$ , since it contains  $\mathcal{R}_t$ . Normalization places  $A_t$  in a hyperplane not containing zero, so  $r_t$  is also the ordinary row rank of the normalized residuals.

If  $R$  is a future array, write  $p_R(a)$  for its next-action probability. It is independent of the next report  $y$ . For  $p_R(a) > 0$ , denote by  $R^{a,y}$  the normalized continuation after that action and report. For an actual residual this is another actual residual at the child cut.

**Proposition 2.1** (Affine slices are closed under positive shifts). *If  $G \in L_t$  and  $p_G(a) > 0$ , then  $G^{a,y} \in L_{t+1}$  for every next report  $y$ .*

*Proof.* Write  $G = \sum_h c_h R_h$  with real coefficients satisfying  $\sum_h c_h = 1$ . Taking an action marginal gives  $p_G(a) = \sum_h c_h p_{R_h}(a)$ . Restriction to  $(a, y)$  is linear in the unnormalized arrays, hence

$$G^{a,y} = \sum_{h: p_{R_h}(a) > 0} \frac{c_h p_{R_h}(a)}{p_G(a)} R_h^{a,y}.$$

The coefficients sum to one. They need not be positive: the display asserts membership in the affine span at the child, not a convex realization. Since  $G \in \mathcal{P}_t$  and  $p_G(a) > 0$ , its normalized shift is a valid nonnegative causal array. Thus it lies in both factors of (3) at the child cut.  $\square$

**Theorem 2.2** (A canonical sufficient criterion and the rank-two case). *If every  $L_t$  is a simplex, a single stochastic clocked machine realizes  $C$  with exactly  $r_t$  labels at cut  $t$ , simultaneously. No realization can use fewer than  $r_t$  at that cut. In particular, every complete finite causal array with  $r_t \leq 2$  at every cut has a simultaneously rank-minimal realization. This includes full-support arrays with arbitrary finite action and report alphabets.*

*Proof.* In any realization, conditioning on a prefix expresses  $R_h$  as a convex combination of the state continuations at cut  $t$ . Its affine dimension therefore forces at least  $r_t$  labels.

Choose the vertices of  $L_t$  as the states at cut  $t$ . Every actual residual is their convex combination. From vertex  $G$ , choose the next action with probability  $p_G(a)$ . After the next report, express  $G^{a,y} \in L_{t+1}$  as a convex combination of the child vertices and use those weights as the update row. An action of probability zero may have an arbitrary unused update. Proposition 2.1 ensures that this construction is possible, and the action probability is independent of the unread report. Initialization uses the barycentric representation of  $C$ . Backward induction verifies the entire specified future law from every vertex state. A simplex of dimension  $r_t - 1$  has exactly  $r_t$  vertices, proving the claim. A zero-dimensional slice is a point and a compact one-dimensional slice is a segment, which proves the last assertion.  $\square$

This criterion differs from requiring the residual hull itself to be a simplex: internal vertices can lie outside that hull. It also differs from searching for unknown enclosing simplices and unknown shifts simultaneously. The polytope  $L_t$  is fixed by linear equations and inequalities from the input array; its normalized-shift closure is proved, not an additional feasibility hypothesis. For fully listed rational arrays, vertices and barycentric rows can be found by rational linear algebra and finite polyhedral enumeration. The explicit array may already be exponentially large in the horizon. No succinct-input polynomial bound, autonomous row-reuse theorem, or universal equality between rank and state complexity follows.

The next section shows that the rank-two conclusion is sharp as a uniform statement about the maximum cut rank. At rank three, separate rank-optimal enclosures need not be causally compatible even for a full-support family.

### 3. A FULL-SUPPORT OBSTRUCTION AT RANK THREE

The following experiment has only two internal memory cuts. The first input is  $x = (x_1, x_2) \in \{-1, 1\}^2$ , received as one symbol. A command  $j \in \{1, 2, 3, 4, 5\}$  is then received, followed by a query  $k \in \{1, 2\}$ . The machine emits  $B \in \{-1, 1\}$  in the update receiving  $k$ . Let  $K_1$  count states after  $x$  and  $K_2$  states after  $j$ ; a retained final answer has two labels. No further access to  $x$  or  $j$  is permitted. For  $0 < h \leq 1$  prescribe the conditional mean vectors

$$(4) \quad \begin{array}{c|ccccc} j & 1 & 2 & 3 & 4 & 5 \\ \hline z_{x,j} & (h, h) & (h, -h) & (-h, 0) & (-h/2, hx_1/4) & (-h/2, hx_2/4) \end{array}.$$

The answer law is  $\mathbb{P}(B = b \mid x, j, k) = (1 + bz_{x,j}(k))/2$ .

The program is clocked and uses arbitrary stochastic rows. Thus every allowed implementation has common rows  $E(s \mid x)$ ,  $T(t \mid s, j)$  and  $D(b \mid t, k)$ ; all past randomness is in  $s$  or  $t$ .

We first record the geometric estimate used by the converse.

**Lemma 3.1** (A quantitative forced triangle). *Put  $h = 1 - \delta$ , where  $0 \leq \delta \leq 1/10$ . If a triangle  $P \subset [-1, 1]^2$  contains  $(h, h)$ ,  $(h, -h)$  and  $(-h, 0)$ , then every point  $(-h/2, y) \in P$  satisfies*

$$(5) \quad |y| \leq \beta(\delta) := \frac{1}{4} + \frac{\delta}{2(1-\delta)} + \frac{\delta}{2-3\delta} < \frac{h}{2}.$$

*Proof.* A convex representation of  $(h, h)$  includes a vertex  $A$  for which  $2 - A_1 - A_2 \leq 2\delta$ , so  $A_1, A_2 \geq 1 - 2\delta$ . A representation of  $(h, -h)$  similarly includes a vertex  $B$  with  $B_1 \geq 1 - 2\delta$  and  $B_2 \leq -1 + 2\delta$ . These vertices are distinct. Call the third vertex  $D$ . In a representation of  $(-h, 0)$ , let  $u$  be the total weight on  $A, B$ . Since their first coordinates are at least  $1 - 2\delta$  and  $D_1 \geq -1$ ,

$$-1 + \delta \geq -1 + 2(1 - \delta)u, \quad u \leq \frac{\delta}{2(1 - \delta)}.$$

The zero second coordinate implies  $|D_2| \leq u/(1 - u) \leq \delta/(2 - 3\delta)$ . For a point of  $P$  with first coordinate  $-h/2$ , the total weight  $v$  on  $A, B$  satisfies

$$-h/2 \geq -1 + 2hv, \quad v \leq \frac{1 - h/2}{2h} = \frac{1}{4} + \frac{\delta}{2h}.$$

Its second coordinate has absolute value at most  $v + (1 - v)|D_2| \leq \beta(\delta)$ . Finally the two fractions in  $\beta$  increase with  $\delta$  while  $h/2$  decreases, and

$$\beta(1/10) = \frac{1}{4} + \frac{1}{18} + \frac{1}{17} = \frac{223}{612} < \frac{9}{20}.$$

This proves the strict bound throughout the interval, including zero.  $\square$

**Lemma 3.2** (Centered square in an inscribed triangle). *If a triangle in  $[-b, b]^2$  contains  $[-a, a]^2$ , with  $a > 0$ , then  $b \geq 2a$ .*

*Proof.* Write its three barycentric functions as  $\lambda_s(u) = c_s + d_s \cdot u$ . They are nonnegative on  $[-a, a]^2$ , so  $c_s \geq a\|d_s\|_1$  and  $\sum_s c_s = 1$ . If the vertices are  $v_s$ , affine reconstruction gives  $\sum_s v_s d_s^T = I_2$ . Taking traces yields  $2 \leq b \sum_s \|d_s\|_1 \leq b/a$ .  $\square$

This necessary bound is sufficient for our contradiction; no claim of its attainability in dimension two is used. Its general-dimensional version belongs to classical cube-absorption geometry; see Section 5.

**Theorem 3.3** (Exact incompatible minima and their Pareto frontier). *For every  $9/10 \leq h \leq 1$ , both cuts of (4) separately admit an exact three-state realization and have ordinary normalized residual rank three. No realization has  $(K_1, K_2) = (3, 3)$ . The entire feasible profile region is*

$$(6) \quad \{(K_1, K_2) : K_1 \geq 3, K_2 \geq 4\} \cup \{(K_1, K_2) : K_1 \geq 4, K_2 \geq 3\}.$$

*Thus the exact Pareto-minimal profiles are  $(3, 4)$  and  $(4, 3)$ , and the minimum full peak is four. For  $h < 1$  the specified array has full support. At  $h = 9/10$  all answer probabilities belong to  $[1/20, 19/20]$ .*

*Proof.* The four rows at the first cut vary independently through the last two second coordinates of (4). They form a nondegenerate affine square, hence have normalized row rank three. At the second cut the response vectors lie in  $\mathbb{R}^2$  and include three affinely independent points  $(h, h), (h, -h), (-h, 0)$ . Its normalized row rank is also three. Factoring through a stochastic register cannot increase rank, so both state counts are at least three.

Here are matching one-sided constructions. Put  $a = h/4$ . At the first cut, three continuation vectors with free coordinate pairs

$$(7) \quad (-a, -a), \quad (3a, -a), \quad (-a, 3a)$$

and the fixed coordinates from (4) enclose the four actual rows. The triangle is legal because  $3a \leq 3/4$ . The barycentric encoder weights for  $(ax_1, ax_2)$  are

$$(\lambda_0, \lambda_1, \lambda_2) = \left( \frac{2 - x_1 - x_2}{4}, \frac{1 + x_1}{4}, \frac{1 + x_2}{4} \right).$$

After  $j$ , represent the resulting two-dimensional mean in the convex hull of the four corners of  $[-1, 1]^2$ . For a mean  $(u, v)$  the weights on  $(\varepsilon_1, \varepsilon_2)$  are  $(1 + \varepsilon_1 u)(1 + \varepsilon_2 v)/4$ . The decoder returns the queried corner coordinate. This is a single machine of profile  $(3, 4)$ .

For profile  $(4, 3)$ , first retain  $x$ . Every actual second-cut mean lies in the triangle with vertices  $h(1, 1), h(1, -1), h(-1, 0)$ . At a mean  $(u, v)$  its weights on those vertices are

$$(8) \quad \frac{h + u + 2v}{4h}, \quad \frac{h + u - 2v}{4h}, \quad \frac{h - u}{2h}.$$

For the last two columns they are respectively  $(1/4, 0, 3/4)$  or  $(0, 1/4, 3/4)$ ; for the first three columns they are the three unit vectors. These rows followed by the binary decoder prove the second construction. All tables are rational when  $h$  is rational.

Suppose now that a  $(3, 3)$  machine exists. Let  $G_1, G_2, G_3$  be its full first-cut continuation arrays. Since their convex hull contains an affine square of dimension two, their affine span equals the affine span of the four actual rows. Therefore each  $G_s$  has the first three columns of (4) unchanged and has last columns  $(-h/2, u_s)$  and  $(-h/2, v_s)$  for some  $u_s, v_s$ . The first encoder then forces

$$(9) \quad [-a, a]^2 \subseteq \text{conv}\{(u_s, v_s) : 1 \leq s \leq 3\}.$$

This affine-slice deduction allows arbitrary state continuations; it neither assumes that they are residuals nor restricts them beforehand to the residual hull.

The three states at the second cut have decoder mean vectors  $D_1, D_2, D_3 \in [-1, 1]^2$ . Their triangle  $P$  contains all actual second-cut rows, and hence the three points in Lemma 3.1. Every shifted continuation of each  $G_s$  must also lie in  $P$ , because the update after  $j$  is a common stochastic row into these same three decoder states. In particular,  $(-h/2, u_s), (-h/2, v_s) \in P$ . Lemma 3.1 gives  $|u_s|, |v_s| \leq \beta(1-h) < h/2 = 2a$ . This contradicts Lemma 3.2 applied to (9). No (3,3) realization exists. Padding the two constructions proves (6). The full-support and probability bounds follow immediately from the displayed means.  $\square$

For  $h < 1$  all residual minimal faces at each cut coincide with the full terminal-channel face. Thus the support-component rank is just three at each cut; support separation is not the source of the failure. The obstruction is simultaneous positivity of the shifted generators. Theorem 2.2 and Theorem 3.3 together locate the first possible maximum cut rank for this phenomenon at three. They also distinguish an actual obstruction from a bilinear feasibility reformulation: the complete feasible profile set is determined here.

#### 4. ADDITIVE ENCODERS ADMIT LOSSLESS ONLINE SYNTHESIS

The preceding obstruction involves two successive nonlinear choices of positive generators. Additivity provides a different situation in which the encoding stage can always be made causal without enlarging its alphabet.

Let  $X_1, \dots, X_n$  be finite input sets and  $S = [m]$  a label set. An encoder is a map  $\lambda : \prod_i X_i \rightarrow \Delta_m$ . Call it additive if there are vectors  $a \in \mathbb{R}^m$  and  $f_i(x_i) \in \mathbb{R}^m$  such that

$$(10) \quad \lambda(x) = a + \sum_{i=1}^n f_i(x_i).$$

The summands need not be probability vectors or have positive entries. All that is assumed is  $\lambda(x) \in \Delta_m$  for every product input.

**Theorem 4.1** (Additive encoding without a persistent selector). *Every encoder (10) has an exact fixed-order streaming realization with at most  $m$  states after each input. This holds for every externally prescribed permutation of the inputs. The construction uses only scalar replacement probabilities and fresh input-dependent draws on  $[m]$ ; it never retains the index of the chosen summand. Rational encoder data give rational transition rows.*

*Proof.* For each label  $s$ , put

$$\ell_{i,s} = \min_{x_i \in X_i} f_{i,s}(x_i), \quad c_s = a_s + \sum_i \ell_{i,s}, \quad g_{i,s}(x_i) = f_{i,s}(x_i) - \ell_{i,s}.$$

Because the input is a full Cartesian product, each scalar minimum can be attained independently. Consequently  $c_s = \min_x \lambda_s(x) \geq 0$ , and  $g_i(x_i)$  is nonnegative. Normalization of (10), with all other coordinates fixed, shows that  $\sum_s f_{i,s}(x_i)$  is independent of  $x_i$ . Therefore

$$w_i = \sum_s g_{i,s}(x_i) \geq 0$$

is constant in  $x_i$ . With  $w_0 = \sum_s c_s$ , one has  $w_0 + \sum_i w_i = 1$ . Delete zero-weight summands. For the others define  $q_0 = c/w_0$  and  $q_i(x_i) = g_i(x_i)/w_i$ , so

$$(11) \quad \lambda(x) = w_0 q_0 + \sum_i w_i q_i(x_i).$$

Let  $W_t = w_0 + \sum_{i \leq t} w_i$ . If  $w_0 > 0$ , initialize from  $q_0$ . At the first positive-weight input when  $W_{t-1} = 0$ , simply draw from  $q_t(x_t)$ ; before it a dummy label suffices. When  $W_t > 0$ , retain the

old label with probability  $W_{t-1}/W_t$  and otherwise replace it by a fresh draw from  $q_t(x_t)$ . A zero-weight input leaves the label unchanged. Induction gives the state law

$$\mathbb{P}(S_t = s \mid x_{1:t}) = \frac{w_0 q_0(s) + \sum_{i \leq t} w_i q_i(s \mid x_i)}{W_t}.$$

At  $t = n$  the denominator is one and the law is exactly  $\lambda(x)$ . Equivalently, each update is the single atomic row

$$(12) \quad T_{t,x_t}(s, s') = \frac{W_{t-1}}{W_t} \mathbf{1}_{\{s=s'\}} + \frac{w_t}{W_t} q_t(s' \mid x_t).$$

Only  $s'$  persists. The weights and their prefix sums are known program data associated with the external clock. The same construction works in every prescribed permutation. Minima and arithmetic preserve rationality.  $\square$

**Corollary 4.2** (Rank-saturating additive channels). *Let  $C_x$  be a finite family of normalized terminal channels, additive in the product coordinates of  $x$ , with affine dimension  $r - 1$ . If  $C_x$  can be represented through  $r$  legal normalized generators at a checkpoint, then it can be acquired sequentially with at most  $r$  labels at every input cut, for any fixed order. Thus its checkpoint and acquisition-peak minima agree at  $r$ . If a final output must be retained, the complete peak is at most  $\max(r, |B|)$  for output alphabet  $B$ .*

*Proof.* A convex representation through  $r$  generators enclosing a set of affine dimension  $r - 1$  makes those generators affinely independent. Their barycentric coordinates are affine functions of  $C_x$ , hence additive in  $x$ . They are stochastic on all product inputs. Apply Theorem 4.1, then the unchanged terminal decoder. The affine-dimension lower bound holds for every acquisition method because the complete input must pass through the checkpoint label.  $\square$

For (1), a decoder state has a mean vector  $v_s \in [-1, 1]^n$ . Conditioning on the state gives

$$(13) \quad \eta x = \sum_s E(s \mid x) v_s.$$

Consequently  $K_n(\eta)$  is the smallest number of outer-cube points whose convex hull contains  $\eta[-1, 1]^n$ . This checkpoint formulation is classical random-access geometry in the normalization of Section 6. An adaptive acquisition schedule still produces one distribution  $E(\cdot \mid x)$ , with no past-dependent input supplied to the decoder, so it obeys the same lower bound.

**Theorem 4.3** (Every simplex encoder has the same optimal online peak). *Let  $n \geq 1$  and  $\eta > 0$ . If a simplex with vertices  $v_0, \dots, v_n \in [-1, 1]^n$  contains  $\eta[-1, 1]^n$ , those very decoder vertices admit an exact online encoder with at most  $n + 1$  states. Hence (2) holds for every  $n$  and  $\eta$ . No hypothesis concerning a Hadamard matrix is required.*

*Proof.* Write the simplex's affine barycentric coordinates as  $\lambda_s(z) = a_s + b_s \cdot z$ . Then  $\lambda_s(\eta x) = a_s + \eta \sum_i b_{s,i} x_i$  is an additive stochastic encoder on the product cube. Theorem 4.1 computes it without expanding the label set. The unchanged binary decoder with mean  $v_{s,j}$  gives (1). Conversely,  $K_n(\eta) \leq W_n^{\text{ad}}(\eta) \leq W_n^{\text{fo}}(\eta)$  and  $K_n(\eta) \geq n + 1$  by affine dimension. A final binary output requires no more than  $n + 1$  labels.  $\square$

The theorem concerns the peak, not simultaneous optimality of the individual  $t + 1$  prefix bounds. Its sampler can occupy all  $n + 1$  labels at an early cut. Nor does it assert online equality for arbitrary nonsimplicial checkpoint codebooks: their barycentric encoders need not be affine. The rank-three example in Section 3 shows why general static factorizations cannot be promoted to joint realizations without additional structure.

### 5. CRITICAL CUBE ABSORPTION AND NON-HADAMARD ONLINE ENCODERS

We separate the classical static geometry from the online consequence. Let  $Q_n = [0, 1]^n$ . For a nondegenerate simplex  $S \subset Q_n$ , let

$$\xi(S) = \inf\{\rho > 0 : Q_n \subset c(S) + \rho(S - c(S))\}, \quad \xi_n = \inf_{S \subset Q_n} \xi(S),$$

where  $c(S)$  is its centroid. The universal lower bound  $\xi_n \geq n$ , the Hadamard equality construction, the axial-diameter formulas and the non-Hadamard equalities  $\xi_5 = 5$ ,  $\xi_9 = 9$  belong to the classical cube-absorption literature; Nevskii and Ukhlov give the needed results and antecedents in [7]. Their Theorem 2 proves that equality in  $S \subset Q \subset nS$ , with homothety about the simplex centroid, forces the centroids to coincide. Thus a critical simplex in their normalization gives, under  $v = 2u - \mathbf{1}$ , precisely

$$(14) \quad n^{-1}[-1, 1]^n \subset \text{conv}\{v_0, \dots, v_n\} \subset [-1, 1]^n.$$

Away from the critical value, centering at the origin and centering at the simplex centroid must not be silently identified.

For reference, the centered obstruction has a short proof. If the barycentric coordinates are  $a_s + b_s \cdot z$ , containment of  $\eta[-1, 1]^n$  gives  $a_s \geq \eta \|b_s\|_1$  and  $\sum_s a_s = 1$ . Reconstruction gives  $\sum_s v_s b_s^T = I_n$ , whence

$$(15) \quad n = \sum_{s,i} v_{s,i} b_{s,i} \leq \sum_s \|b_s\|_1 \leq \eta^{-1}.$$

This is a proof of a classical inequality in the present notation, not a new static threshold. The extra conclusion is that every feasible simplex, critical or not, has the same online peak by Theorem 4.3.

**Proposition 5.1** (Explicit online rows for a critical simplex). *For any simplex satisfying (14), write its barycentric functions as  $a_s + \sum_i b_{s,i} z_i$ . Then*

$$(16) \quad a_s = \frac{1}{n+1} = \frac{1}{n} \sum_i |b_{s,i}|, \quad \sum_s |b_{s,i}| = 1, \quad \sum_s b_{s,i} = 0.$$

*The distributions*

$$(17) \quad q_i(s \mid x_i) = |b_{s,i}| + b_{s,i} x_i$$

*and updates*

$$(18) \quad T_{t,x_t}(s, s') = \frac{t-1}{t} \mathbf{1}_{\{s=s'\}} + \frac{1}{t} q_t(s' \mid x_t)$$

realize the critical query channel with  $n+1$  states. Multiplying the final decoder means by  $n\eta$  gives every  $0 < \eta \leq 1/n$  at the same peak. The origin of a replacement draw is not retained.

*Proof.* Every inequality in (15) is an equality at  $\eta = 1/n$ . Thus  $a_s = n^{-1} \|b_s\|_1$  for each  $s$ , and  $v_{s,i} b_{s,i} = |b_{s,i}|$  for each pair  $(s, i)$ . Evaluating the  $s$ th barycentric function at its own vertex gives  $1 = a_s + \|b_s\|_1 = (n+1)a_s$ , so the centroid is zero. The  $i$ th diagonal entry of  $\sum_s v_s b_s^T = I_n$  yields the second identity; barycentric normalization yields the third. Hence (17) is a probability vector. Induction in (18) gives

$$\mathbb{P}(S_n = s \mid x) = \frac{1}{n} \sum_i q_i(s \mid x_i) = a_s + \frac{1}{n} \sum_i b_{s,i} x_i.$$

This is exactly the barycentric distribution of  $x/n$ . Its decoder mean is  $x_j/n$ , and the claimed attenuation completes the proof. Unlike a Hadamard construction, an individual  $q_i$  need not have mean vector  $x_i e_i$ ; the required cancellation holds after averaging all coordinates.  $\square$

**5.1. Two rational non-Hadamard implementations.** The following vertices are taken from [7, Theorem 4 and Section 9], in its  $[0, 1]^n$  normalization. They are reproduced to specify the actual online rows, not as new static constructions. In dimension five the six rows of  $U_5$  are

$$(19) \quad U_5 = \begin{pmatrix} 1/2 & 1 & 1/3 & 1 & 1 \\ 1/2 & 0 & 1/3 & 1 & 1 \\ 1/2 & 1/2 & 1/3 & 0 & 1 \\ 1/2 & 1/2 & 0 & 1/3 & 0 \\ 0 & 1/2 & 1 & 1/3 & 0 \\ 1 & 1/2 & 1 & 1/3 & 0 \end{pmatrix}.$$

In dimension nine, choose the parameter  $t = 1/2$  in their family:

$$(20) \quad U_9 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1/2 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1/2 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1/2 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1/2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1/2 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1/2 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1/2 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1/2 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Put  $V_d = 2U_d - \mathbf{1}$  entrywise and  $M_d = [\mathbf{1} \ V_d]$ . The inverse gives  $\lambda_s(z) = (1, z^T)(M_d^{-1})_{\cdot s}$ ; its first row is  $a_s$  and its other rows are  $b_{s,i}$ . Direct rational multiplication gives

$$(21) \quad \begin{array}{c|ccc} d & |\det M_d| & a_s & \sum_i |b_{s,i}| \\ \hline 5 & 32 & 1/6 & 5/6 \\ 9 & 12800 & 1/10 & 9/10 \end{array}$$

and  $V_d^T(b_{s,i})_{s,i} = I_d$ ,  $V_d^T a = 0$ . Thus the displayed matrices also verify (14) directly: all barycentric coordinates are nonnegative on  $d^{-1}[-1, 1]^d$ , sum to one and reconstruct  $z$ . Equations (17)–(18) are full transition formulas requiring only these finite rational inputs.

**Corollary 5.2** (Online optimality beyond Hadamard orders). *For  $d = 5$  and  $d = 9$  and every  $\eta > 0$ ,*

$$K_d(\eta) = W_d^{\text{ad}}(\eta) = W_d^{\text{fo}}(\eta) = d + 1 \iff \eta \leq 1/d.$$

*The matrices (19)–(20) and (18) give rational acquisition rows. No Hadamard matrix of order six or ten is used or needed.*

*Proof.* Apply Proposition 5.1 to the two matrices, and use the dimension lower bound and (15) for the converse.  $\square$

In the terminology of [7], the five-dimensional seed is perfect; their nine-dimensional seed is critical but is not perfect. Equality of the absorption factor, not the stronger condition that every cube vertex touches a dilated face, is what the online argument uses.

**5.2. Closure of normalized critical matrices.** This elementary closure gives an explicit growing family of such encoders. Let  $M = [\mathbf{1} \ V]$  be the augmented matrix of a centered critical simplex of dimension  $d$ , and let  $m = d + 1$ . Its centroid condition and (16) say that the first row of  $M^{-1}$  is  $\mathbf{1}^T/m$  and every column of  $M^{-1}$  has absolute sum one. Conversely, these two conditions together with  $V \in [-1, 1]^{m \times d}$  and invertibility imply the critical inclusion by barycentric positivity.

**Proposition 5.3** (Normalized tensor closure). *If  $M$  and  $N$  satisfy the preceding conditions, so does  $M \otimes N$  with its all-ones column placed first. It gives a critical simplex of dimension  $(d+1)(e+1) - 1$ . Thus tensor products of the rational seeds of sizes 2, 6 and 10 give explicitly specified optimal online simplex encoders whenever*

$$n+1 = 2^a 6^b 10^c > 1, \quad a, b, c \text{ nonnegative integers.}$$

*Proof.* Entries of  $M \otimes N$  are products of entries in  $[-1, 1]$ , so the new vertices are in the cube. The first inverse row is uniform on the product index set. Since  $(M \otimes N)^{-1} = M^{-1} \otimes N^{-1}$ , the absolute column sums multiply and remain one. Invertibility and the converse observation prove the critical inclusion; the online assertion follows from Proposition 5.1. The seed of size two is  $\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ .  $\square$

No general multiplicativity of nonnegative rank is asserted. This is an explicit operation on the normalized critical matrices just defined; the static tensor algebra is not used as a priority claim.

## 6. EXACT QUERY SYNTHESIS AND CLASSICAL WORST-CASE CODING

Fix a finite decoder dictionary  $D = \{v_1, \dots, v_m\} \subset [-1, 1]^n$ , with private encoder and decoder randomness and no shared seed. A worst-case success guarantee  $(1 + \eta)/2$  means that for every input  $x$  there is a probability vector  $E_x$  such that

$$x_i \sum_s E_x(s) v_{s,i} \geq \eta \quad \text{for all } i.$$

The same  $E_x$  must serve all future queries. Merely requiring average success over queries is a different condition.

**Proposition 6.1** (Checkpoint exactification with no added states). *For  $0 \leq \eta \leq 1$  the following are equivalent for the fixed dictionary:*

- (1) *it achieves worst-case random-access success at least  $(1 + \eta)/2$ ;*
- (2)  $\eta[-1, 1]^n \subseteq \text{conv } D$ ;
- (3) *it exactly realizes (1) after changing only the encoder.*

*Thus the minimum checkpoint alphabet in the exact problem equals the minimum alphabet in the private-randomness worst-case coding problem. No runtime symmetrization or retained random partition is necessary.*

*Proof.* Assume the first condition and fix  $u \in \mathbb{R}^n$ . Choose an input corner with  $x_i = \text{sign}(u_i)$  when  $u_i \neq 0$ , and let  $z_x = \sum_s E_x(s) v_s$ . Then

$$\max_{v \in \text{conv } D} u \cdot v \geq u \cdot z_x \geq \eta \sum_i |u_i|.$$

The right side is the support function of  $\eta[-1, 1]^n$ . The separating hyperplane theorem gives the second condition. It gives an exact convex representation of every  $\eta x$  using the same dictionary, which is the required new encoder in the third condition. The third condition implies the first immediately.  $\square$

This is a translation of the geometry already present in [6, Theorem 14, Lemma 15 and equations (143)–(149)], not a new coding exponent. In their  $[0, 1]^n$  normalization the dictionary is  $S = (D + \mathbf{1})/2$ , its directed  $\ell^\infty$  Hausdorff error from the Boolean corners is  $e(D)$ , and the largest centered cube factor is

$$\eta(D) = 1 - 2e(D), \quad p_{\text{worst}}(D) = 1 - e(D) = \frac{1 + \eta(D)}{2}$$

whenever the factor is nonnegative. The factor two is essential when passing from errors to means. Their model has  $m = 2^k$  messages; the proposition uses any integer  $m$ . Their Lemma 15 explicitly passes to cube containment. It is therefore not appropriate to call the exact checkpoint channel a strictly harder cardinality problem solely because it specifies equality rather than a lower bound on success.

Exactification can change a non-affine encoder into a different non-affine encoder. The proposition does not bound the storage needed to compute it online. Theorem 4.1 supplies that missing step for additive encoders, and Theorem 4.3 supplies it for every simplex dictionary. This is the precise division between the static coding theorem and the online theorem here.

The first-order entropy rate  $1 - h_2((1 + \eta)/2)$  and logarithmic communication overhead are classical [1]. The preceding revision proved a fully charged blockwise exact realization with logarithmic state excess  $O_\eta(\sqrt{n \log n})$ , and uniform row-error results; those proofs and statements remain in the unchanged supporting article. Neither exactification nor the present online simplex theorem proves a matching second-order law for that block construction. The retained entropy theorem is not used as a new contribution of this revision.

Likewise, the optimal  $(2^k - 1, k)$  classical construction and its static bound in [6, Theorems 16–17] overlap the Hadamard/Sylvester checkpoint case. The online synthesis applies to its simplex without increasing the number of messages, but does not change that existing optimality statement. Their conjecture-dependent results for other message sizes remain conjecture-dependent; none is used to infer an unconditional finite phase diagram here.

## 7. FINITE COINS AND STOCHASTIC-LANGUAGE CONVENTIONS

**7.1. A separate fair-bit register ledger.** The exact label counts above refer to atomic stochastic rows. They do not include the internal computation of a real-probability sampler. For the rational constructions a fully finite implementation is available, with an additional cost which can be stated without hiding a counter or the current input.

**Proposition 7.1** (Finite fair-bit implementation of rational rows). *Consider  $T$  input stages, an at-most- $m$ -label transducer with  $m \geq 2$ , and current input alphabet size at most  $d$ . Suppose every row has a common denominator for its entries of size at most  $D \geq 2$ . Put  $b = \lceil \log_2 D \rceil$ . In a self-paced protocol which supplies the next input only after the preceding update completes, the transducer has an exact fair-bit implementation using at most*

$$(22) \quad 2(T + 1)md2^b$$

*states, including phase, current input, old label, sampler prefix and terminal output. Each stage uses fewer than  $2b$  fair bits on average. Thus its persistent bit cost is at most*

$$\lceil \log_2(T + 1) \rceil + \lceil \log_2 m \rceil + \lceil \log_2 d \rceil + b + 2.$$

*This changes the implementation model and does not assert preservation of the atomic optimum in total states.*

*Proof.* For a row of denominator  $D_\rho \leq D$ , let  $b_\rho = \lceil \log_2 D_\rho \rceil$ . Draw  $b_\rho$  fair bits to form an integer  $J$  between zero and  $2^{b_\rho} - 1$ . If  $J \geq D_\rho$ , repeat; otherwise map  $J$  through the row's integer cumulative sums to the new label. A denominator-one row is deterministic. The acceptance probability is  $D_\rho/2^{b_\rho} > 1/2$ , unless it is one, and the accepted integer is uniform. This proves exactness and the expectation bound.

Retain the phase, old label and current input throughout an attempted draw. The proper binary prefixes of length less than  $b_\rho$  form a tree with  $2^{b_\rho} - 1 \leq 2^b - 1$  nodes. On the last bit, acceptance can move directly to the ready state containing the new label; rejection returns to the root. Hence a separate leaf or unbounded trial counter is unnecessary. There are at most

$(T + 1)m$  ready states,  $Tmd(2^b - 1)$  sampler states and two final states. Their sum is at most (22). The input provider waits at an explicit ready boundary; the phase is stored, not supplied as a free data-dependent clock.  $\square$

For (19) the distributions  $q_i$  have denominators dividing six, and for (20) they divide ten. Thus a row at acquisition stage  $t$  has denominator dividing  $6t$  or  $10t$ , respectively. The critical decoders are also rational. More generally the normalized matrix tensors in Proposition 5.3 have rational denominators bounded by products of fixed seed denominators; their logarithms are  $O(\log(n+1))$ . The finite-coin implementation of those families therefore uses  $O(\log(n+1))$  persistent bits, with explicit rational row data. This assertion concerns the critical polynomial-size matrix families, not efficient construction of a fixed-interior-signal exponential codebook.

## 7.2. Internal normalization of probabilistic automata.

**Definition 7.2** (Proper stochastic-language automaton). On a finite alphabet  $\Sigma$ , a proper probabilistic automaton consists of a finite state set, an initial probability row  $\alpha$ , nonnegative transition matrices  $T_a$  for  $a \in \Sigma$ , and stopping probabilities  $\tau_s$ , with

$$\tau_s + \sum_{a,t} T_a(s, t) = 1.$$

Stopping emits no further symbol; a transition emits its label  $a$ . Termination is almost sure from every reachable state. The generated stochastic language and state languages are

$$L(w) = \alpha T_w \tau, \quad L_s(w) = e_s^T T_w \tau.$$

They are probability distributions on finite words. Unreachable states are removed. The automaton is residual if every reachable state language is  $L_u(w) = L(uw)/L(u\Sigma^*)$  for some positive-mass prefix  $u$ .

These are the conventions used when comparing with probabilistic residual automata [3, 2]. If a nonnegative representation initially has state continuation masses  $h_s \in (0, \infty)$ , normalization replaces  $T_a(s, t)$  by  $T_a(s, t)h_t/h_s$ ,  $\tau_s$  by  $\tau_s/h_s$ , and  $\alpha_s$  by  $\alpha_s h_s$ . The identity  $h_s = \tau_s + \sum_{a,t} T_a(s, t)h_t$  gives stochastic rows. States with  $h_s = 0$  have no contribution to any generated word and are deleted. Thus finite nonzero continuation masses can be normalized without adding states. Infinite continuation mass is not a stochastic language and is not included in this convention.

For completeness, we retain the operational separation with a proof in these internal terms. It is a result of the preceding revision, not a new succinctness-priority assertion.

**Proposition 7.3** (A normalized residual/positive separation). *For  $n \geq 2$  and rational  $0 < \eta \leq (2n - 1)^{-1}$ , define the finite language*

$$(23) \quad L(u_{x_1} \cdots u_{x_n} q_j o_b) = \frac{2^{-n}}{n} \frac{1 + b\eta x_j}{2},$$

*using disjoint input, query and output symbols, and give zero mass to all other words. Every residual automaton has at least  $2^n$  states. A proper rational positive automaton has at most  $(n + 1)(n + 2)/2 + n(n + 1) + 1$  states, including phase and query storage.*

*Proof.* The query-phase residuals are the  $2^n$  vertices of an affine cube, with probabilities  $(1 + b\eta x_j)/(2n)$  on the two-symbol words  $q_j o_b$ . Conditioning any proper automaton on a supported prefix expresses that residual as a convex combination of normalized state languages. Nonnegativity forces each used language to be supported on those two-symbol words. A residual of (23) after length  $t$  is concentrated on length  $n + 2 - t$ . Hence a residual state used here must itself be one of the query-phase cube vertices. Extremality forces a separate state for each of the  $2^n$  vertices.

For the upper bound, generate the input bits independently and fairly. At phase  $t$  use labels  $0, 1, \dots, t$ . On the next input, retain the old label with probability  $t/(t+1)$ ; otherwise replace it by  $t+1$  for a positive bit and zero for a negative bit. After  $t$  inputs the probability of label  $i > 0$  is  $\mathbf{1}_{\{x_i=1\}}/t$  and the probability of zero is the fraction of negative bits. At phase  $n$ , associate means  $v_0 = -\eta \mathbf{1}$  and  $v_i = -\eta \mathbf{1} + 2n\eta e_i$ . They are in the cube when  $\eta \leq (2n-1)^{-1}$ , and their mixture has mean  $\eta x$ . Emit  $q_j$  uniformly, store  $(s, j)$ , emit  $o_b$  with mean  $v_{s,j}$ , and terminate. Phase tags require  $\sum_{t=0}^n (t+1)$  states, query buffers require  $n(n+1)$ , and one state terminates. Every state language is normalized and every row is rational. Multiplication of the three stages yields (23).  $\square$

## 8. CONSEQUENCES AND RETAINED RESULTS

The main distinction is now witnessed, rather than merely formulated. At full-support cut rank three, independent minimum positive factorizations need not fit one machine; the exact profile set (6) exhibits the cost of that incompatibility. At cut rank at most two, canonical affine sections always yield compatible minimum realizations. Additive encoders give a different structural class with no online alphabet penalty, regardless of the size or shape of the original residual hull.

For the binary-query problem, the minimal  $n+1$  checkpoint alphabet has an online realization whenever it exists. The critical geometric threshold is therefore shared by checkpoint and full acquisition peak, not only in Hadamard dimensions. The static absorption results are classical; the transition formulas transfer each of them to an actual online machine. At larger message alphabets a generic encoder need not be additive, and the present theorem does not identify every finite state count or the second-order streaming excess. No new complete varying-signal phase diagram is inferred.

The full-support two-cut experiment has a fixed order of three atomic inputs. It establishes a compatibility obstruction, not a theorem optimizing all informative experimental actions or all report-dependent collision-validation schedules. The earlier controlled minimax, marked-saddle, positive-noise collision, fixed-clock scheduling, revelation, entropy-rate and acquired-transport results remain in the unchanged supporting article and cumulative mathematical volumes. Their statistical experiments and resource conventions are not replaced by the binary-query interface. Known-target exact stochastic rows and acquired-target approximate guarantees remain distinct.

The repository's historical A2/A3/A4 and B2/B1/B3/B4/C1/C2/D1 derivations have independent Fourier, large-deviation, operator-domain, semigroup and optional-projection hypotheses. Preserving those derivations does not make them consequences of a finite positive-realization theorem. The new proof dependencies are internal and explicit: affine slices locate the rank boundary; the forced-triangle obstruction proves a nonattainable profile; additive synthesis transfers critical simplices to online memory; exactification identifies the correct classical checkpoint comparison. No historical analytic gate is marked complete on the strength of that transfer.

The primary literature comparison also has precise limits. We use the cube-absorption statements and explicit simplices of [7], and the full v3 worst-case coding characterization of [6]. Positive realization and residual-state representations are older frameworks [5, 10, 3, 2]; their general existence language is not claimed as an invention. The support/affine arguments here must be distinguished from arbitrary nonnegative-rank minimization [4], and complete arrays from incomplete machine specifications [9]. General filtered-experiment comparison [8] is not a priority claim of the paper. The exact quantifiers and explicit constructions, rather than the size of the preserved development, are the mathematical assertions submitted for independent examination.

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